

PEGINTERFERON BETA-1A -> immunodeficiency 45

For clinical-collaborator outreach.

Why this hypothesis is real

Drug-target convergence on the disease scores **0.62** (noisy-OR over OT target-disease associations). This pair is fully novel relative to all known drug-indication assignments in ChEMBL. Therapeutic direction is **aligned** with OT genetic evidence (the drug pushes the target the disease-protective way). Lead target is well expressed in the disease-relevant tissue (blood). Orthologous-gene model-organism evidence is strong (phylo_score 0.55, n=1). Reactome co-membership reinforces the indirect mechanism with **3** shared specific pathways.

Why now (IP runway + literature footprint)

The substance is a biologic or otherwise outside the FDA Orange Book; check the Purple Book and direct registry for IP runway.

Clinical opportunity

US prevalence figures are not in the curated map; for an orphan condition this is expected. The KOL can size the addressable population.

What we propose

BioBix (Department of Mathematical Modelling, Statistics and Bioinformatics, Ghent University) brings bioinformatics support to the collaboration:

- Target validation across blood / immune / cancer multi-omics datasets
- Biomarker candidate identification from public + private cohorts
- Patient stratification analytics for trial-enrichment design
- Mechanism-of-action mining via systems-biology priors
- Quick-turn evidence syntheses for IIT and partnered programs

Proposed collaboration model:

- Investigator-initiated trial framework with the substance owner
- BioBix as bioinformatics analytics partner (target/biomarker/biostatistics)
- Joint publication strategy; data-sharing agreement gated on IP terms

Next step

A 30-minute scoping call to align on feasibility. We would come prepared with a brief mechanistic dossier, an annotated literature pull, and a pilot biomarker shortlist for the disease.

Wim Van Criekeinghe, Professor (PhD, h-index 73)

Department of Mathematical Modelling, Statistics and Bioinformatics, Ghent University

Wim.VanCriekeinghe@UGent.be